Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 2: Power series solutions to ODEs

How Topic 2 builds on Topic 1:

- The solutions to the ODEs in Topic 1 involve familiar "elementary" functions that we have seen countless times in pre-requisite subjects (e.g., polynomials, exponentials, trigonometric etc)
- In Topic 2, we will encounter a richer class of ODEs whose solutions cannot be expressed in terms of these elementary functions. Instead they give rise to power series solutions which are classed under "Special Functions".

Topic 2: Power series solutions to ODEs

Leaning outcomes:

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1. Discover how power series and associated ODEs arise in real-world applications

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2. Learn power series method for solving ODEs and IVPs

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1. Discover how power series and associated ODEs arise in real-world applications
2. Learn power series method for solving ODEs and IVPs
3. Where does a given series converge? To which function does a series converge?

Lecture 7

We will:

1. Revise the concept of a power series and calculate power series for some simple functions
2. Motivate the use of power series for solving ODEs
3. Revise the ratio test for determining the radius of convergence of a power series

Topic 2: Power Series Solutions to ODEs

- You are familiar with the concept of a series, e.g.,

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how to test if they converge and what they converge to.

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- The series may be functions of t, e.g., a power series

$$\sum_{n=0}^{\infty} c_n (t-a)^n$$

Definition (Taylor series)

- If $f(t)$ is N-times differentiable, the Nth-order Taylor polynomial around $t = a$ is

$$P_N(t) = \sum_{n=0}^{N} \frac{f^{(n)}(a)}{n!} (t-a)^n$$

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- If $a = 0$ then it is often known as a Maclaurin series

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$$P(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} = 1 + t + \frac{1}{2}t^2 + \ldots$$

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$$f(t) = \frac{1}{1-t} \qquad \Rightarrow \qquad f(0) = 1$$

$$f(t) = (1-t)^{-1}$$

$$= u^{-1} \qquad \| \; u = 1-t$$

$$\frac{df}{dt} = \frac{df}{du} \frac{du}{dt}$$

$$= -u^{-2} \times -1 = u^{-2} = (1-t)^{-2}$$

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Use this to find the Taylor series

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$$f^{(n)}(t) = \frac{n!}{(1-t)^{n+1}} \qquad \Rightarrow \qquad f^{(n)}(0) = n!$$

$$P(t) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(t-0)^n = \sum_{n=0}^{\infty} \frac{n!}{n!} t^n = \sum_{n=0}^{\infty} t^n$$

As this is a geometric series it converges for $|t| < 1$

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$\ldots$ what if the exponential function hadn't been discovered?

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Try a ~~Maclaurin~~ power series of form

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$$\Rightarrow \quad y'(t) = \sum_{n=1}^{\infty} n\, c_n\, t^{n-1}$$

making an assumption!

$$\frac{d}{dt} \sum_{n=0}^{\infty} c_n t^n = \sum_{n=0}^{\infty} \frac{d}{dt} c_n t^n =$$

Notes on this slide were made after lecture in response to question

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$$\Rightarrow \quad \underbrace{\sum_{m=0}^{\infty} (m+1)\, c_{m+1}\, t^m}_{y'} = \underbrace{\sum_{m=0}^{\infty} c_m\, t^m}_{y} \quad \Rightarrow \quad \sum_{m=0}^{\infty} \Big\{ (m+1)\, c_{m+1} - c_m \Big\}\, t^m = 0$$

Therefore, require

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Let $c_0 = C$, then

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Convergence of power series

Theorem (Ratio test for power series)

- Consider the power series

$$P(t) = \sum_{n=0}^{\infty} c_n \, (t-a)^n$$

where $c_n \neq 0$ for all large n.

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- The power series converges if
$$L = \lim_{n\to\infty} \left| \frac{c_{n+1}(t-a)^{n+1}}{c_n(t-a)^n} \right| = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n}(t-a) \right| < 1$$
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Need to use other methods for when $L = 1$ as the ratio test is inconclusive for this case.

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Thus, the radius of convergence is one

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Therefore, the series converges for $-1 \leqslant t < 1$.

Exercise 2.5: Use the ratio test on the series

$$P(t) = \sum_{n=1}^{\infty} c_n \, t^n \quad \text{where} \quad c_n = \begin{cases} \frac{1}{2^n} & (n \text{ even}) \\ 0 & (n \text{ odd}) \end{cases}$$

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i.e., every other term is zero.

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Apply ratio test to sequential non-zero terms $n = 2\,k$ and $2\,(k+1)$

$$\frac{c_{2(k+1)}\, t^{2(k+1)}}{c_{2k}\, t^{2k}}$$

$$\lim_{k\to\infty} \left| \frac{t^{2(k+1)}}{2^{2(k+1)}} \frac{2^{2k}}{t^{2k}} \right| < 1$$

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Apply ratio test to sequential non-zero terms $n = 2\,k$ and $2\,(k+1)$

$$\lim_{k \to \infty} \left| \frac{t^{2(\cancel{k+1})}}{2^{2(\cancel{k+1})}} \frac{\cancel{2^{2k}}}{\cancel{t^{2k}}} \right| < 1$$

$$\Rightarrow \quad \left| \frac{t^2}{4} \right| < 1 \quad \Rightarrow \quad |t| < 2$$

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$$\lim_{k \to \infty} \left| \frac{t^{2(k+1)}}{2^{2(k+1)}} \frac{2^{2k}}{t^{2k}} \right| < 1$$

$$\Rightarrow \quad \left| \frac{t^2}{4} \right| < 1 \quad \Rightarrow \quad |t| < 2$$

Could also have rewritten the series as $P(t) = \displaystyle\sum_{k=1}^{\infty} \left(\frac{t}{2} \right)^{2k}$

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- $P(2) = \sum_{k=1}^{\infty} \frac{2^{2k}}{2^{2k}} = \sum_{k=1}^{\infty} 1$, which diverges

Now testing the endpoints

- Note that that $P(-2) = P(2)$
- $P(2) = \sum_{k=1}^{\infty} \frac{2^{2k}}{2^{2k}} = \sum_{k=1}^{\infty} 1$, which diverges

Thus, $P(t)$ converges for $-2 < t < 2$.